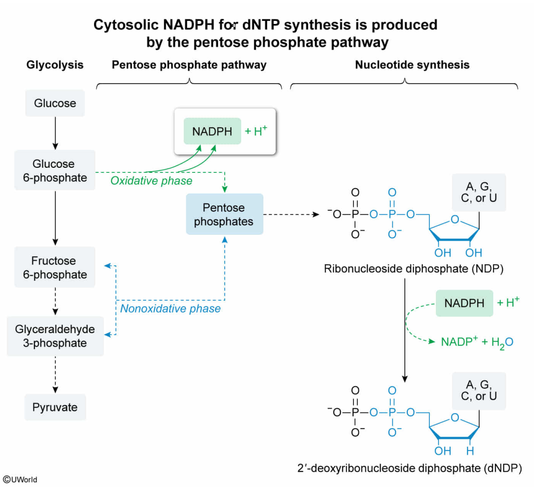
The cytosolic production of deoxyribonucleotides prior to DNA replication requires the reduction of ribonucleoside diphosphates in a multistep mechanism that involves free radicals and the ultimate use of NADPH as a reducing agent. The cellular store of this reducing agent is produced primarily from:

- A. the citric acid cycle. [11%]
- B. glycolysis. [3%]
- C. fatty acid metabolism. [5%]
- ✔ D. the pentose phosphate pathway. [78%]

Correct
77% answered correctly

Explanation:



Oxidation–reduction reactions are an important class of biochemical reaction. Many biochemical redox reactions are linked to energy production—for example, the NADH or $FADH_2$ produced in glycolysis and the citric acid cycle feed into the electron transport chain, where the energy of their high-energy electrons is used to produce ATP. Other redox reactions are linked to synthesis reactions (eg, fatty acids, cholesterol) or protection against oxidative damage; these reactions may use other redox cofactors like **NADPH**.

NADPH is a reducing agent produced during the oxidative phase of the **pentose phosphate pathway**. During the oxidative phase, two molecules of NADPH are produced over three enzymatic steps as glucose 6-phosphate is oxidatively decarboxylated to a pentose phosphate. The NADPH produced in this pathway is the **primary cytosolic source of NADPH in mammalian cells**.

The question introduces another synthesis reaction involving redox—the **reduction of nucleotides to deoxynucleotides**. The question states that this pathway uses NADPH as the ultimate reducing agent. Because the question states that this pathway is **cytosolic**, the NADPH necessary for nucleotide reduction comes **primarily from the pentose phosphate pathway**.
(Choices A and B) The citric acid cycle and glycolysis both produce the reducing agent NADH, *not* NADPH. Furthermore, the NADH (as well as the $FADH_2$) produced by the citric acid cycle are produced in the mitochondria, *not* the cytosol.

(Choice C) Fatty acid oxidation produces NADH and $FADH_2$ (*not* NADPH) and also occurs in the mitochondria rather than the cytosol. Fatty acid synthesis occurs in the cytosol but *consumes* NADPH; it is not a *source* of NADPH.

Educational objective:
NADPH is a reducing agent used in various biosynthetic pathways and in protecting the cell against oxidative damage. The primary cytosolic source of NADPH production is the oxidative phase of the pentose phosphate pathway.

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Subject: Biochemistry	**System:** Metabolic Reactions

A cell line is treated with a novel electron transport chain inhibitor suspected to inhibit Complex III. In theory, when mitochondria are treated with this inhibitor, researchers would be able to detect increased levels of:

I. Ubiquinone

II. Oxidized cyt *c*

III. NAD^+

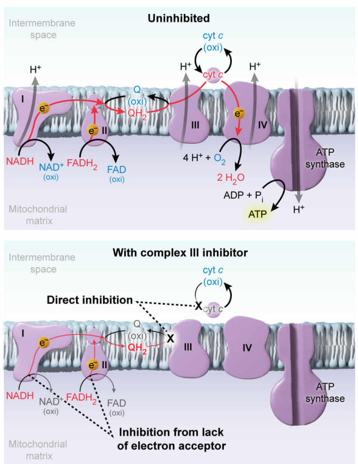
IV. FAD

✔ ◉ A. II only [40%]
○ B. III only [9%]
○ C. I, III, and IV only [38%]
○ D. I, II, III, and IV [12%]

Correct
41% answered correctly

Explanation:

Inhibition of Complex III leads to accumulation of oxidized cytochrome *c*



Red = reduced cofactor; blue = oxidized cofactor.

The electron transport chain (ETC) is a series of protein complexes in the inner mitochondrial membrane that couples the movement of electrons through a series of redox reactions to the **pumping of protons** across the inner membrane. This forms a proton gradient, which is then used to synthesize ATP during cellular respiration. The ETC consists of four complexes:

- Complex I receives a **pair of electrons** from **NADH**. The electrons are passed through a series of intermediate redox centers within the complex before **reducing ubiquinone to ubiquinol**.

- Complex II receives a pair of electrons from **$FADH_2$**. Like Complex I, the electrons

  ultimately **reduce ubiquinone to ubiquinol**.

- Complex III receives electrons from the **ubiquinol** produced by either Complex I or Complex II and ultimately transfers them to **cytochrome *c* (cyt *c*)**. For every pair of electrons received, two cyt *c* are reduced.

- Complex IV receives electrons **from cyt *c*** and transfers them to molecular **oxygen (O_2)**. Four electrons are needed to fully reduce a molecule of oxygen, so the stoichiometry per reduced cofactor (ie, NADH or $FADH_2$) is two electrons reduce half a molecule of O_2, producing one molecule of H_2O.

The question states that the inhibitor **inhibits Complex III**; therefore, the inhibitor **directly prevents cyt *c* from being reduced**. Complex IV is still active and can continue to receive electrons from any pre-existing reduced cyt *c*, resulting in an **accumulation of oxidized cyt *c* (Number II)**.

(Number I) With Complex III inhibited, the reduced ubiquinol will not be oxidized back to ubiquinone. This leads to an accumulation of *reduced* ubiquinol, not oxidized ubiquinone.

(Numbers III and IV) Because Complex III cannot regenerate oxidized ubiquinone, no electron acceptors are available for NADH and $FADH_2$ to transfer their electrons. Therefore, the *reduced* forms (NADH and $FADH_2$) will accumulate, not the oxidized forms (NAD^+ and FAD).

Educational objective:
The electron transport chain (ETC) is a series of inner mitochondrial membrane protein complexes that transfers electrons from molecule to molecule during cellular respiration. If an ETC complex is inhibited, reduced cofactors will accumulate prior to the point of inhibition, and oxidized cofactors will accumulate after the point of inhibition.

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Subject:
Biochemistry

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System:
Metabolic Reactions